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# Climate Change

In climate change firstly we discuss about climate

## Climate:

"Weather over a long period"

## Climate Changes:

Climate change refers to long-term shifts in temperature and weather patterns.

## Explanation:

● Climate change in the world can be caused by various activities.

● When climate changes occur temperature can increase. When temperature ises many different changes.

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occur on Earth.

Climate change result in  
More

• Flood  
↓  
due to melting of glaciers

• Sea Heat Waves  
↓  
Oceans are warming  
and becoming more acidic  
and disturb Aquatic life.

• Droughts  
↓  
a lack of precipitation  
over a long period of time.

• Intense Rain  
also leading the Floods

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## ● Introduction:-

Natural causes of climate change are processes and phenomena that occur without human influence. They factors have caused fluctuations in Earth's climate for million of years. These sources include:-

● Green house Gas emission

● Solar Variation

● Volcanoes

● Earth orbital change

● Ocean Currents



# Causes of climate change



1) Continental drift

2) Volcanoes

3) The Earth's Tilts

4) Ocean Currents

5) Intensity of Solar

Radiation

1) Green Houses Gases

- Carbon dioxide ( $\text{CO}_2$ )

- Methane ( $\text{CH}_4$ )

- Nitrous oxide ( $\text{NO}_2$ )

- Chloro fluoro carbons (CFCs)

- Ozone ( $\text{O}_3$ )

- Water Vapors ( $\text{H}_2\text{O}$ )

2) Land Use Change

- Deforestation

- Urbanization

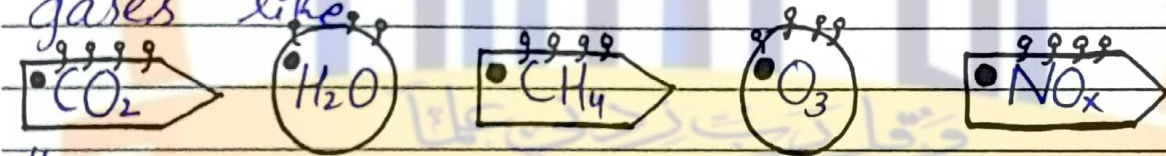


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# Natural Green House Gas Emission

Natural greenhouse gas emission have always played a key role in shaping Earth's climate. These natural emissions from various sources and release gases like,



These gases trap the incoming radiations coming from the Sun and hence cause climate change.

Here are some natural sources to emission of these gases:-

Volcanoes release a significant amount of  $\text{CO}_2$  during eruption

Respiration  
Plants, animals and micro-organisms release  $\text{CO}_2$  during cellular respiration



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Natural wildfires

Burning of organic material like forest releases  $\text{CO}_2$

Wetlands:-

Decomposition of organic matter in water-logged soil produces  $\text{CH}_4$ .

Microbial process

Denitrification by bacteria in soil produces  $\text{N}_2\text{O}$

Evaporation and Transpiration:-

water enters atmosphere through evaporation. Plants release water in transpiration.

## 2:- Solar Radiation Variation

○ Changes in Sun's Energy output can affect Earth's Climate-

○ Periods of high energy such as sunspot cycles, can lead to warming, while lower activity can cause cooling.

## 3:- Volcanic Eruption:-

Volcanic eruptions release vast amount of ash, sulphur dioxide, Carbon dioxide and many other gases into the

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atmosphere. Emission of  $\text{CO}_2$  gas contribute to gradual warming.

## 4: Ocean Currents:-

Ocean currents are continuous, directed movement of sea water driven by various factors acting upon water.

- Transport warm water from equator towards poles and cold water towards equator and thus regulates global temperature.
- Fluctuations in ocean currents leads to extreme weather events like
  - Droughts
  - Floods
  - Hurricanes

## 5- Earth Orbital Change:-

It refers to the periodic variation in Earth's orbit and orientation related to sun. These changes influence the amount of solar radiation, which in turn affect long-term climate change.



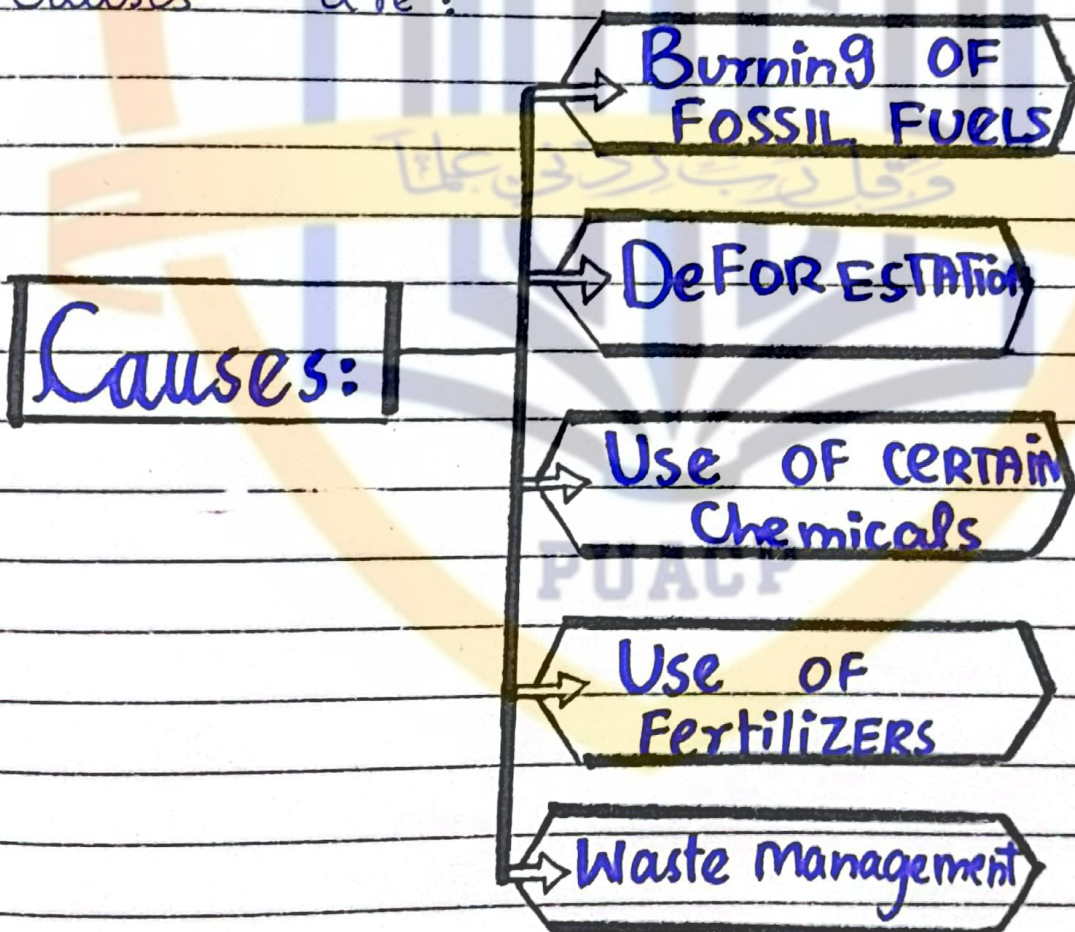
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# Man-made Causes of Climate Change:-

Man made causes of climate change comes from **activities of humans** that increase greenhouse gases in the atmosphere. Main Causes are:





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## Burning of Fossil Fuels:-

The combustion of coal, oil and natural gas for energy releases certain amounts of  $(CO_2)$  and other greenhouse gases into the atmosphere. Which then cause an increase in **global warming**.

## Deforestation:-

Cutting down of forests for agriculture, urban development and logging reduces the no. of trees that can absorb  $CO_2$ . This **increase concentration of  $CO_2$**  in the atmosphere. As it is greenhouse gas and causes **global warming**.



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## Use of Certain Chemicals:-

The use of certain chemicals such as Chlorofluorocarbons (CFCs) and hydrofluorocarbon (HFCs) in refrigeration, air conditioning and aerosols contribute to the depletion of Ozone layer and global warming.

## Use of Fertilizers:-

Fertilizers used for crops especially rich in nitrogen releases Nitrous Oxide, a greenhouse gas, during decomposition in the soil.

## Waste Management:-

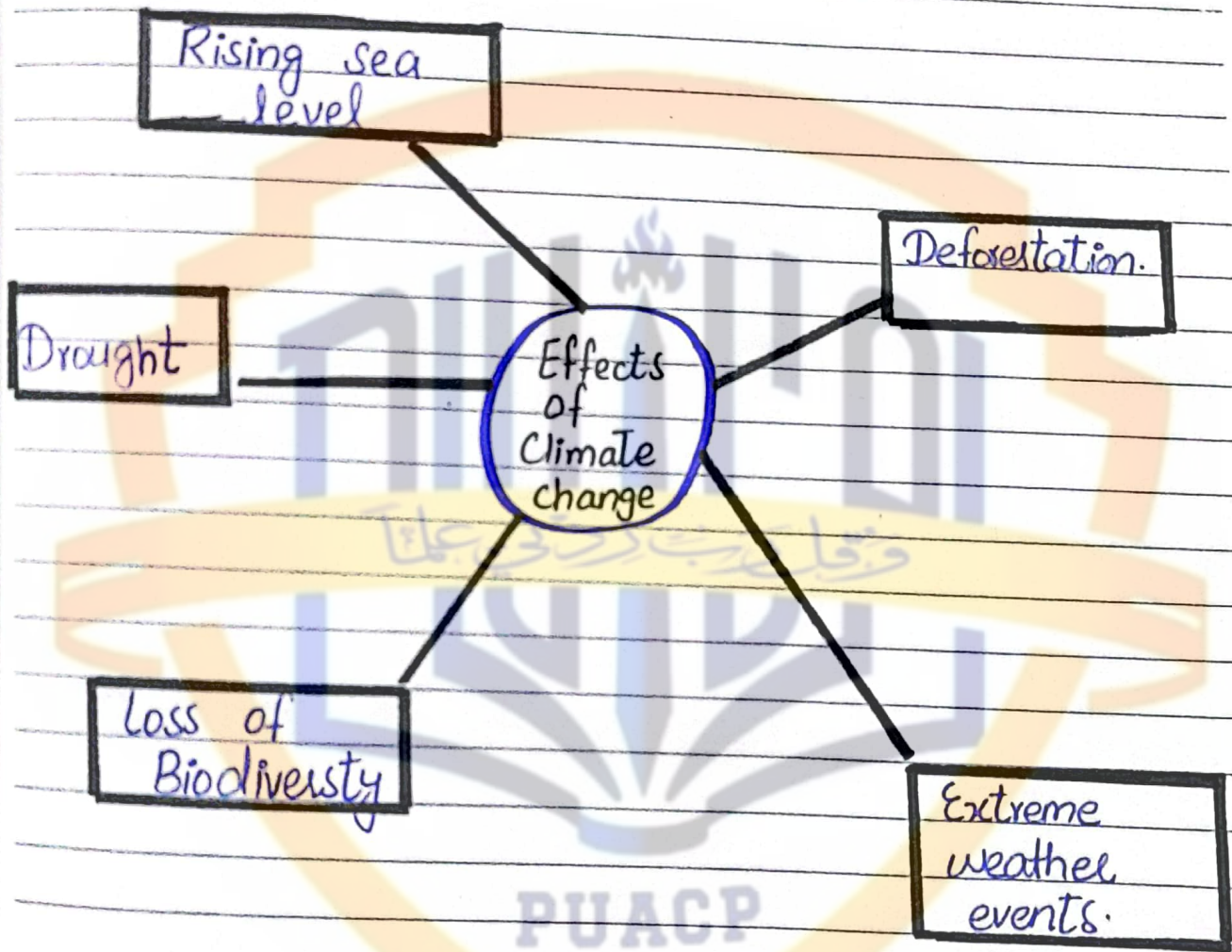
Livestock waste often decomposes in an organAerobic environment which produces the methane ( $\text{CH}_4$ ) gas, a potent greenhouse gas.



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## Effects of Climate Change:-





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## 1. Rising sea level:-

Thawing of polar ice-caps and glaciers, leading to coastal erosion and flooding. A sea level rise is an increase in global mean sea level as a result of an increase in the volume of water in the world's ocean.

## 2. Extreme Weather Events:-

Increased frequency and intensity of heat waves, droughts, storms and heavy rainfall. Climate change is increasing the frequency and intensity of extreme weather events.

## 3. Loss of Biodiversity:-

Climate change can effect loss of biodiversity. Climate changes alter ecosystems, leading to extinction of plants and animals. Ecosystems are constantly changing.



#### 4. Deforestation:-

Deforestation is the permanent destruction of forest. Rising temperature and changing precipitation patterns lead to forest fires and degradation. When carbon stored by trees is released back into the atmosphere as carbon dioxide, which contributes climate change.

#### 5. Drought:-

A long period of low rainfall, which affects growing and living conditions. Any unusual dry period which results in a shortage of water. A drought can last for several years or months. Lakes, rivers and streams may as well run dry.

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# Preventive Measures

Invest In  
renewable  
energy sources

Improve farming  
and vegan diets

Ending our reliance  
on fossil fuels

Protect forests and  
plants

Switch to sustainable  
transport



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## 1. Invest In Renewable Energy Sources:

- We should use renewable energy sources like sun, wind, water and heat.
- By using these source, there will occur improvement in our environment because they emit no green house gases.

## 2. Improve Vegan Diet:

- By shifting diets from meat and animal products to plant based diets.
- There will occur reduction in the carbon element of environment.

## 3. Ending Reliance On Fossil Fuel:

- By reducing the use of fossil fuels, reduction ultimately occur in the production of hazardous gases, which effect our climate.

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#### 4. Protection Of Forests and Plants:

- Afforestation and Reforestation activities play a vital role for maintaining change in climate.

#### 5. Sustainable Transport:

- By using and promoting collective transport such as trains, public buses etc. There will be reduction occur in the emission of gases that effect our environment. We should have to use electric vehicles.

By adopting all these preventive measures, improvement will occur in climate change.

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